

TranqAuricle: A novel, portable and non-invasive device that detects anxiety, records its frequency and calms the brain through the auricle.

Synopsis:

A WHO study report, conducted for the NCMH, states that 6.5% of Indian population suffers from some form of serious mental disorder. Current drugs available to treat anxiety disorder bind to large areas of the brain, leading to unwanted side effects.

TranqAuricle is a novel, portable and non-invasive device that detects anxiety, records its frequency and calms the brain through the auricle. It uses an integrated circuit holding a pulse sensor, Lm35 temperature sensor, a vibrating motor, a motor driver and a microcontroller (Arduino nano). The device is attached to the ear such that the pulse sensor is placed at ear tips (i.e. directly above veins). The pulse sensor and Lm35 temperature sensor provide analog input to the microcontroller which in turn produces output if it receives fluctuations in heart beat rate and skin temperature respectively. This directs the vibrating motor to generate vibrations into a menthol gel which spreads, producing a cooling, relaxing effect across ear lobes. This causes the nerve impulses to travel to the cerebral cortex where the emotional triggers dampen and anxiety is sharply reduced. Further, it electronically keeps a track on the frequency of panic attacks, to be accessed in future.

TranqAuricle has no measurable side-effects. It helps the patients in self-recovery and alleviates uneasiness of opening up before psychologists. It simply needs to be put across the ear. It is cost-effective and non-invasive.

Introduction:

Characterized by extreme feelings and emotions of prolonged worry and sense of dread, anxiety is the most prevalent mental health condition with as many as 10 million cases per year in India. Despite their widespread occurrence, these disorders have not received the same recognition as other major syndromes. Anxiety disorders account for decreased productivity, increased suicidal rates, and rise in alcoholism and drug abuse. Addiction to antidepressants and sedatives rise among the sufferers. Discomfort of opening about personal life during cognitive behavioral therapies further worsens such disorders.

Objectives:

- Breaks the unhealthy thought-feeling-action cycles by calming the anxious brains, automatically.
- Records the frequency of panic attacks, measure stress levels and store the data in a memory chip which can be accessed in future through any IOT platform.
- It helps anxiety patients in self-recovery and averting the discomfort of opening up before psychologists.
- Allows the nerve impulse to pass to cerebral cortex or consciousness section of the brain where the person is least anxious.
- Avert ghastly consequences of depression, panic attacks, anxiety and reduce suicidal rates.

Innovation:

The project is a handy and miniature ear device that pacifies the apprehensive brains by generating vibrations into a menthol gel that spreads, producing a cooling and relaxing effect across ear lobes. They contain vital nerves linked to brain and heart and are made up of tissue rather than cartilage unlike other ear parts, thereby making it one of the most sensitive parts of the body. Any impact on this part, will affect the whole body and hormonal secretion. TranqAuricle detects anxiety dually, through rapid fluctuations of temperature in the ear lobe and beating heart rate. It also keeps a track on the frequency of panic attacks, electronically which can be retrieved and used in future. When the brain calms down, the nerve impulses travel to the conscious section of the brain (cerebral cortex) where analytical and rational thinking takes place, replacing negative emotions.

There have been various other projects such as Pacifica, Headspace, Muse, emWave2, Stress thermometers and many more. But all previous gadgets focus on engaging users with meditation, calming exercises, games and involve mere detection of mood swings and stress. Moreover neurostimulation therapies, release excess of dopamine and serotonin resulting in nausea and fatigue. In TranqAuricle, anxiety is controlled without the use of any expensive setups or mobile apps or human intervention. The innovative part is that this single system performs 3 major tasks concerned with the disorder, simultaneously.

TranqAuricle is a one way solution to the mind-numbing disorder, anxiety.

Algorithm:

Sequence of steps followed:

- First, we conducted a local survey in Cosmopolis and DN OxyPark Apartment, Bhubaneswar and took 35 people under observation.
- We collected newspaper articles, national and international statistics highlighting the increasing trends of anxiety and other mental ailments.
- Then, we visited experts in the field of neurology and psychology enquiring about anxiety assessments, measuring scales for anxiety and cognitive behavioral therapies.
- We visited VLCC institute and learnt about the benefits of massaging during anxiety or stress.
- In the ATL Lab of our school, we began programming and assembling our device.
- We then collected the materials required for cooling gel experiment and recorded the data.
- In a self-made domestic laboratory, we prepared menthol gel sachet as a part of the device.
- Lastly, we carried our survey collecting data and reviews of our project, from prestigious medical institutions of Bhubaneswar.

Methods:

- **LOCAL SURVEY** - We surveyed 35 people in Cosmopolis and DN OxyPark Apartment on the basis of a questionnaire assessment. The questionnaire included reasons, common symptoms, effects of anxiety, different methods of coping and their age. We prepared a data list in accordance to the survey and interpreted it through bar graphs and pie charts.

- **MEDICAL AND VLCC INSTITUTE SURVEY** – We enquired and collected data about the drawbacks of anxiety medications and current therapies, benefits of vibrational massage and menthol in our body, and discussed future scopes of the project.
- **ATL LAB**—We began Arduino programming for the Lm35 temperature sensor, pulse sensor and vibrator. We assembled the parts and miniaturized the device in our school ATL lab.
- **COOLING GEL EXPERIMENT**-First, we added 7gm of propylene glycol to 3gm of menthol crystals in a test tube and shook it properly. Later it was placed in a small tub of hot water for breaking down the remaining crystals. We compared the cooling effect and evaporation time of menthol to that of ethanol and normal water. The data was then recorded.

Conclusion/Results:

- **LOCAL SURVEY-**
 1. MOST COMMON REASON OF ANXIETY: PHOBIAS OR FEARS
 2. MOST COMMON EFFECT OF ANXIETY: LOSS OF CONCENTRATION
 3. MOST COMMON AGE GROUP SUFFERING FROM ANXIETY: 15-25 YEARS
 4. MOST COMMON SYMPTOM OF ANXIETY: INCREASE IN HEARTBEAT
 5. MOST COMMON WAY OF COPING ANXIETY: MEDITATION
- **MEDICAL INSTITUTE SURVEY**
 1. Drawbacks of current therapy – Compliance are the major drawbacks of these therapies along with social stigma that is associated with “mental disorders”.
 2. Drawbacks of current medication – Homeopathic medicines have side effects like dizziness, headache, and difficulty in swallowing and even breathing problems.
 3. Most common symptom of anxiety- Increase in heartbeat and fluctuation in body temp.
 4. Vibrational massage specifically is capable to lower the stress hormone (cortisol) in the body.
 5. Menthol triggers the cold-sensitive TRPM8 receptors in the skin and increases GABAergic transmission in PAG neurons responsible for reducing neuronal excitability throughout the CNS.
- **ATL SURVEY**
 1. Manually the heartbeat was increased (by running or jumping) and the pulse sensor was tested which gave a result of increased heartbeat rate.
 2. With the help of external factors, the body temperature was increased and Lm35 sensor was tested which gave an output of increased skin temperature.
 3. When both pulse sensor and Lm35 temperature sensor provide analog input to the microcontroller it produces output directing the vibrating motor to generate vibrations.

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